

Deklen Crocker

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EDUCATION

BSc. in Applied Science of Electrical Engineering

Edmonton, AB | Nov 2022

UNIVERSITY OF ALBERTA

Coursework: Digital Circuit Design, Digital Logic, Embedded System Design, IC Design

WORK EXPERIENCE

JONES MICROWAVE INC. | EMBEDDED SYSTEM DEVELOPER

Edmonton, AB | Sept 2022 – Present

- Design custom laser driver circuits to interface with bleeding-edge RF components.
- Interface with microwave and optical engineer to develop custom RF switches and attenuators.
- Create custom PCB designs using KiCAD for ECAD and FreeCAD for MCAD Design requirements
- Develop unique testing plans for the custom PCBs using oscilloscopes, function generators, optical sensors to confirm design is within spec.
- Collaborate with other teams to complete thorough reporting on work completed over project milestones.
- Add comment on Organizational work (Contributions to IT, HR, etc.)
- Add comment on circuit design with LTspice

GOVT. OF THE NORTHWEST TERRITORIES | PROJECTS INTERN

Inuvik, NT | Apr 2021 - Aug 2021

- Build Microsoft Visual Basic for Application tools to streamline project information from engineers to Projects Manager.
- Compile financial and status reports from 100+ for the Projects Manager to present to the Deputy Minister for Infrastructure.
- Audit projects with contractor disputes to ensure infrastructure projects are completed on time and within budget.

GOVT. OF THE NORTHWEST TERRITORIES | PROJECTS INTERN

Inuvik, NT | Apr 2020 - Aug 2020

- Developed a Personal Health Record (PHR) system in **Spring** based on the theoretical model outlined in [Mitu Kumar's thesis](#).
- Applied the mCP-ABE encryption scheme using the JPBC library, so patients have fine-grained access control over their health records with the ability to instantly revoke access.

PROJECTS

CLASSROOM IOT CARBON DIOXIDE SENSOR (CAPSTONE)

KiCAD, DART, ESP32, FLUTTER

A security research tool to send targeted spam messages on either Facebook or Twitter. The program analyzes the feed of its target to increase the effectiveness of the messages.

IMAGE REPOSITORY

JAVA, PHP, REACT, TYPESCRIPT, MAVEN, GCP, MYSQL

A full-stack image repository, where users can store their files. Created a GCP cloud function to implement CP-ABE encryption and used it to ensure files are secure at rest.

SKILLS

Languages: Verilog, C++, Python, Perl, SQL

CAD: FreeCAD, KiCAD, Fusion, Shapr3D

Technology: Git, AWS, GCP, Azure, Docker, Unity, Apache, $\LaTeX$, MongoDB, DynamoDB, Neo4j